AIRSTREET · CONCEPT OPTIONS

AirStreet concept options

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	Three options with different working principles, each modelled to detail level D3 in cad/src/model.py and scored against AST-URS-001 (kit 1.9.1 upgrade); the current design is option A; recommendation proposed, awaiting Amish

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Result: option A, the current design (an electrochemical NO2 cell and a laser-scattering PM sensor in a naturally ventilated pod on FieldNode's solar core), scores 221 of 260; option C, passive diffusion tubes with a battery PM logger, scores 198 (10.4 % below A); option B, an aspirated, heated-inlet sampling node, scores 188 (14.9 % below A). No option is within 10 % of A, so the recommendation is A, proposed, awaiting Amish (AST-DEC-001, open decision 1; AST-DDR-004). The design is not changed: A stays the design of record, as STANDARDS section 21 requires for a repo designed before kit 1.8. The best idea of the runner-up, C, is worth grafting onto A: triplicate diffusion tubes beside some nodes as a cheap, independent check of the annual NO2 means (open decision 2). All three options are modelled to detail level D3 in `cad/src/model.py` (Amish, 2026-10-24: "Detail every option"), and the options picture below is rendered from those models by `python .kit/options.py`. Method: STANDARDS section 21.

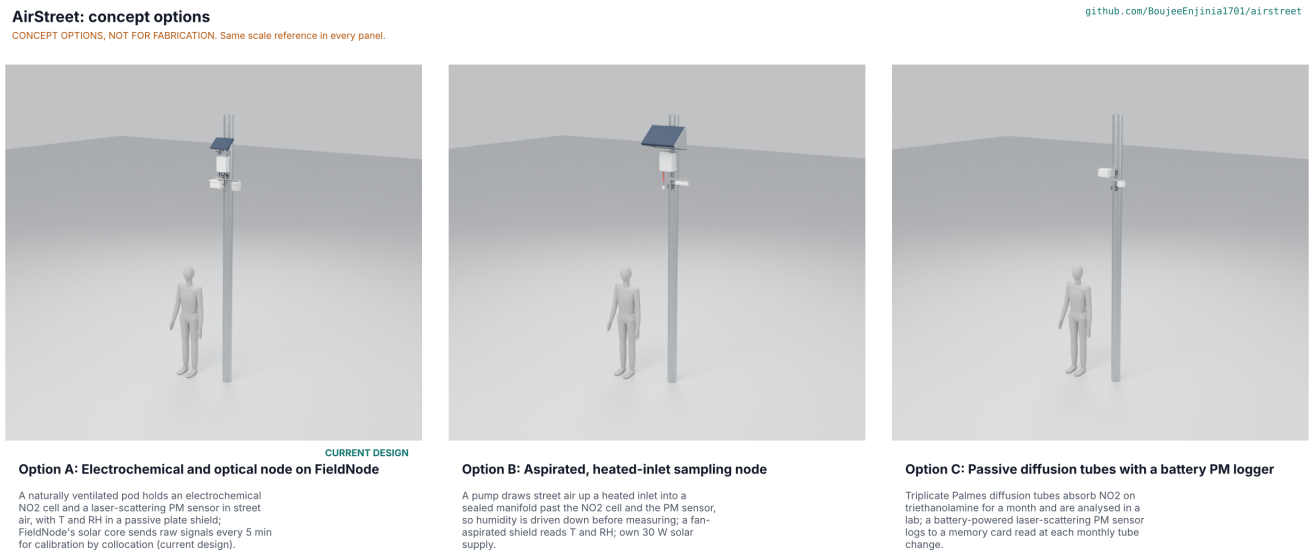


Figure 1. The three options on the same 140 mm street light pole with a 1.75 m person for scale (Blender Cycles render of the D3 models; CONCEPT OPTIONS, NOT FOR FABRICATION). The picture is rendered on Amish's Mac and is not yet in the repository.

The options

All three mount on the same pole with the same printed 140° V-saddles and cut-to-length stainless bands on a 40 x 4 mm aluminium rail, put their inlets 3.0 m above the sidewalk, and send no images, audio or personal data. They differ in how the air reaches the sensors and how NO2 is measured.

Option A: electrochemical and optical node on FieldNode (current design)

A naturally ventilated printed pod hangs from a cross arm under the FieldNode core. Street air enters from below through insect mesh: a laser-scattering particle sensor (SPS30 class) draws its own sample for 60 s every 5 min, and a four-electrode electrochemical NO2 cell with an ozone filter (NO2-B43F class) faces down through the pod floor, read by a potentiostat and a 16-bit ADC. Temperature and humidity are read in an eight-plate passive shield. FieldNode's 6 W panel and 6 Ah LiFePO4 cell power it, and raw signals go out by LoRaWAN every 5 min for calibration by field collocation with a reference station. Risks: humidity above 85 % (PR-001), field NO2 accuracy

unproven (PR-002), sensors outside their rated range on hot afternoons and humid nights (flagged, AST-DDR-005 F2; validity for Amish, AST-STM-001 S33). Needs from a partner: a reference station to collocate at, the pole owner's permission, the radio band.

Option B: aspirated, heated-inlet sampling node

A small diaphragm pump draws about 1 L/min of street air up a 10 mm stainless inlet under a rain cap, through a 2 W heater sleeve that warms the sample a few kelvin so its humidity falls below 80 %, into a machined manifold where the NO2 cell sits in the flow path, and on to the particle sensor in a sealed printed flow box. A 40 mm fan draws air through a double-tube shield for T and RH. With the pump, heater and fan the node needs about 28 Wh a day, so it carries its own 30 W panel, a 12 Ah LiFePO4 pack and a controller in an IP65 enclosure on a back plate. It answers the humidity weakness of A at the cost of power, size and parts. Risks: a heated inlet evaporates part of the nitrate in the particles (AST-STM-001 B4); pump wear; a winter energy deficit in cloudy climates. Needs from a partner: the same as A, plus a site with room for a 0.2 m² panel.

Option C: passive diffusion tubes with a battery PM logger

Three Palmes diffusion tubes hang open end down under a printed hood on a cross arm that spaces them more than 50 mm from the pole. Each tube holds steel grids coated with triethanolamine at its closed end; NO2 diffuses up the 71 mm tube and is absorbed, and after two to four weeks the tubes are capped and sent to a laboratory, which reports a monthly mean. Beside it an IP65 box holds a particle sensor behind a louvred cowl, a logger board with a memory card and an SHT45, and four D cells, read and changed at each monthly tube change. No radio, no solar. This is how UK local authorities map street NO2 (Defra guidance), so its calibration route is well established, but it gives monthly NO2 only and needs a visit at height every month. Risks: no hourly data; theft or loss of tubes; monthly labour. Needs from a partner: an accredited analysis laboratory and a reference station for bias adjustment.

First-order numbers

Table 1. Options compared. Option A figures from AST-CAL-001 v0.7 and the BOM; B and C are first-order estimates for comparison only. Parts and fidelity masses from `cad/fidelity.json` (the kit's densities: printed ASA and the circuit boards are taken as HDPE, the FieldNode internals as solids, so these masses are indicative and differ from AST-CAL-001).

Quantity	Option A	Option B	Option C
Estimated parts cost (USD)	302.50 sensor head (priced BOM); 433.50 with the FieldNode core	About 500 for the whole node (estimate)	About 130 for the hardware (estimate), plus about £8 per tube with analysis (Mapping for Change, 2015 price), 36 tubes a year per site
Mass on the pole (kg)	3.61 (3.77 with FieldNode's sun shield), AST-CAL-001	About 6 (model)	About 1.5 (model)
Frontal area (m²)	0.139	About 0.25, the panel dominates (estimate)	About 0.06 (estimate)
Parts in the D3 model (with fasteners)	195 (103 fasteners)	148 (104 fasteners)	78 (49 fasteners)
Fidelity mass (kg)	3.3	5.9	1.5
NO2 data	Hourly, about 4.3 ppb MAE after field calibration (at risk)	Hourly, similar cell, steadier humidity	Monthly mean, about 25 % uncertainty
PM2.5 data	Hourly; at risk above 85 % RH	Hourly; dried sample, nitrate loss	Logged every 15 min, read monthly; no shield for RH
Energy	2.33 Wh/day on FieldNode	About 28 Wh/day (estimate), 30 W panel	About 0.5 Wh/day, four D cells a month
Service	Pod exchange every 6 months, about 14 min	Pod, filters and pump every 6 months (estimate)	Every month, tubes and cells
STEM check	Open: four fixes made, eleven rows plausible, one (S33, validity above 40 degC) for Amish	B4 plausible (nitrate loss)	Within known science

Scoring

Table 2. Weighted scoring against the user requirements of AST-URS-001 (High weighs 3, Medium 2, Low 1) and the four general criteria; scores 1 to 5.

Criterion	Source	Weight	A	B	C	Reason
Fine particles on my street	UR-001	3	4	5	3	B dries the sample; A at risk above 85 % RH; C has no shielded RH for the correction
Traffic NO2 on my street	UR-002	3	4	4	3	A and B use the same cell; C gives monthly means only
Rank streets with a stated uncertainty	UR-003	3	4	4	3	Hourly data with a stated budget against monthly means at about 25 %
Data often enough for street maps	UR-004	3	5	5	2	C reports NO2 monthly and PM only when the card is read
A calibration others can check	UR-005	3	5	5	5	Collocation route in A and B; C follows the established bias adjustment of the Defra guidance
Raw data to reprocess	UR-006	2	5	5	3	C's NO2 is a laboratory result, not a raw signal
Fits existing poles without harm	UR-007	3	5	4	5	B's larger panel and box weigh more on the same clamps
Small and light on the pole	UR-008	3	4	1	5	A 3.77 kg against 4.0 kg; B about 6 kg and 0.25 m²; C about 1.5 kg
Works outdoors all year	UR-009	3	3	4	4	A's sensors are flagged at the hot and humid ends; B conditions the sample; C's tubes are simple
Quick, simple service at height	UR-010	2	4	3	2	A every 6 months; B adds pump and filters; C a visit every month
Nothing personal collected	UR-011	3	5	5	5	Levels only in all three
Affordable for a district network	UR-012	3	3	2	4	From Table 1; C is cheap to build but each tube costs about £8 to analyse
Own power supply	UR-013	3	5	3	4	B needs a 30 W panel and has a winter margin to prove; C swaps cells monthly
Open and buildable	UR-014	2	5	3	5	B needs a machined manifold and a pump
Safe to install and service	UR-015	3	4	3	3	B is heavier with a heater circuit; C means twelve climbs a year
Radio within the rules	UR-016	2	5	5	5	A and B met on paper; C has no radio
Cost against the value-engineering target		2	3	2	4	A is USD 17.50 over the USD 285 target for the sensor head
Buildable with local skills and tools		2	4	3	5	
Safety		3	4	3	3	As UR-015
Maintenance		1	4	2	3	
Weighted total			**221**	**188**	**198**	

Weighted total = sum of score x weight over the 20 criteria (maximum 260). C is 10.4 % below A and B 14.9 % below A, so no option is within 10 % of the leader.

Recommendation

Proposed, awaiting Amish: carry option A forward as the design of record (AST-DEC-001, open decision 1; argued in AST-DDR-004). A is the only option that gives hourly NO2 and PM2.5 inside the pole owner's load and on the lab's shared solar core. C wins on mass, cost to build and simplicity and is a respected method for annual NO2, but it cannot show rush hours or daily patterns (UR-004) and needs a monthly climb. B answers A's humidity weakness but fails the load and area limits by a wide margin.

The runner-up's best idea to graft onto A: triplicate diffusion tubes under a small hood on the same cross arm at a few nodes, changed every two to four weeks by a partner as the Defra guidance asks, as a cheap independent check of the annual NO2 means and of the street transfer bias that limits PR-004 (open decision 2). From B, a short heated section at the PM inlet, switched on only above 80 % RH, is the idea to test if PR-001 stays at risk after collocation; it costs power that FieldNode's budget does not have today, so it is a note, not a proposal.